

Fractions on Strips and Number Lines



Grade 3 • Mathematics
NC.3.NF.2

Connect equal parts with fraction notation.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Each part is $\frac{1}{2}$. 2 copies make $\frac{2}{2}$. The point is 2 equal steps of size $\frac{1}{2}$ from zero.



Build a second model together

A strip is divided into 2 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Discuss: What should the first step find?



Your turn • 1

A strip is divided into 4 equal parts. 1 part is shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Represent → solve → explain → check



Your turn • 2

A strip is divided into 8 equal parts. 6 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Represent → solve → explain → check



Your turn • 3

A strip is divided into 6 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Represent → solve → explain → check



Your turn • 4

A strip is divided into 2 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Represent → solve → explain → check



Exit question • Explain independently

A strip is divided into 4 equal parts. 2 parts are shaded. Write the shaded fraction. Mark the same quantity on a number line from 0 to 1.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 1/4

Exit: 2/4

2. 6/8

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. 2/6

4. 2/2

